

Manoj Kumar Gudala

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Career Highlights:

- 11 Years + of Experience in Storage and Embedded systems domains, emphasis on Programming and Design.
- Having Experience as a Firmware Engineer in storage Products
- Strong programming skills in C and Assembly programming languages.
- Hands on Experience on firmware level programming for 8 bit, 16bit, 32 bit Micro-controllers and its hardware interfacing.
- Experience in Storage Environments such as SAN/NAS.
- Good understanding of SCSI, SAS, SATA.
- Good understanding of Lustre File Systems
- Experience on Hardware designing and testing.
- Experience with full software life cycle specification, design, implementation, integration, debug, and maintenance
- Willing to work with customer from design to production
- Good understanding of operating systems and computer architecture.

Training & Education Details:

- Master of Engineering in Embedded System Technologies from VelTech College, Anna University, Chennai with an aggregate of 79.4%.
- Bachelors of Engineering in Electronics & Communication Engineering from Saveetha Engineering College, Anna University, Chennai with an aggregate of 68.33% .
- Certified in Advanced Embedded systems course from Vector India, Hyderabad.

Professional Experience:

<i>Company</i>	<i>From – To</i>	<i>Designation</i>
Micron India PVT LTD	May 2019-till date	Staff Firmware Engineer
Gyansamhitha Tech Pvt LTD (Client: ATP electronics Taiwan)	May 2018-May2019	Senior Developer
Seagate India HDD Pvt Ltd	Nov 2017- April 2018	Senior Engineer
MSys Technologies (Client: Seagate India HDD PVT LTD)	Dec 2016- Nov 2017	Senior Engineer
Oracle India Private Ltd, Hyderabad	2013 Jul to Dec 2016	Software Developer
Vedams Software Solutions, Hyderabad.	Jul 2010 to Jul 2013	Software Engineer
RVR Institute of Engineering & Technology	Jun 2009 to May 2010	Asst Professor (ECE)

Technical Skills:

Programming languages : C, C++, Embedded C, Assembly Language,
Protocols : NAS, FTP, I2C.SPI, CAN.
Operating Systems : Linux (SUSE, REDHAT), VxWorks, Windows 2000/XP/7
Processors/Controllers: 8051, Pic16F877, Atmel, ARM
Tools : Wire Shark, Val grind, GDB. Keil, Slick editor.
Configuration Tools : Git

Projects at Micron India

Working as Staff firmware engineer on MAND Products for new controller architectures.
Involved in developing ROM code & fixing the issues found in DV testing

Project Name:

Micron MNAND

Domain: ROM

Environment: C.

Description:

MNAND focus on developing NAND storage for both mobile & auto motive business industry. Where we are involved in design and implementation of ROM Firmware code various new controllers.

Role & Responsibilities:

- Implementation of Test framework for both simulation and emulation environment.
- Implemented new features as per the requirements
- Fixed issues raised in design verification.

Project Name:

Micron Client SSD

Domain: SSD- Frontend Environment

Environment: C.

Description:

This is an SSD in house drive that is end to end developed by Micron on top of the PCIe 3.0 with NVMe 1.3 compliance. Worked on Micron client-based model SSD controller firmware development & Responsible in developing the features like Sanitize, Power optimization, Thermal management features & optimization. Involved in critical bug fixes raised in FTE/PVE & SQRA test environments.

Role & Responsibilities:

- Handled the Admin commands & VS commands support in the SSD firmware from FE side.
- Fixed the issue in Boot flow during the firmware initialization & DDR initializations flow understanding.
- Fixed multiple timing & core issues which are involved in Power state Transitions (PS0/PS3/PS4).

- Implementation of Sanitize feature from FE perspective & involved in BUG fixes with the FTL team on the same.
- Implemented end to end design of improving the NAND thermal profiling with less impact on IO.
- Handling different types of resets in the NVMe.
- Providing trainings & helping validation team in understanding the issues.

Projects at Gyansamhitha Tech Pvt Ltd- India

Worked as Senior developer on SSD Products for ATP electronics products. Working on SM2258 controller (Silicon Motion) SSD controller. Responsible in developing the code to configure the RAM and FLASH Chips on the board and configuring the module in ROM (read only memory) mode.

Project Name:

MC code loader.

Domain: SSD

Environment: C.

Description:

Design the flow of operations to be performed on a factory manufacturing SSD disk and bring it into production mode. This involves configuring the DRAM and FLASH chips available on the board. Verify the FLASH chipset mapping and detecting the bad blocks and providing all the configuration values to the next stage.

Project Name:

ROM Boot & Boot Main code.

Domain: SSD

Environment: C.

Description:

Design the chip Boot Mode from the scratch and configure the SATA path to accept basic vendor commands to boot the main firmware. This involves reading the flash for the main firmware saved in it. Also, this code handles the DEVSLP (dev sleep) functionality.

Project Name:

Test power & mode control automation.

Domain: SSD

Environment: C.

Description:

In the testing process we observed we need to power off-on the device, automatically & remotely. Also, we need set the board in BOOT mode or MAIN code mode. This was initially done through manual intervention. However this process couldn't help us in make test cases execution auto and run remotely. This problem was overcome with my idea of controlling the board power and mode pins with an Arduino controller.

Role & Responsibilities

- ✧ Developing the code from initial flow.
- ✧ Involved in adding new features and improvements in the code.

- ⤴ Handling issues and providing the bug fixes observed during testing.
- ⤴ Adding new ideas to simplify the test process.

Projects at Seagate India

Working as Senior Engineer on Storage Products for DotHill products (Formally acquired by Seagate). Responsible in doing analysis to root cause the DU (data unavailability) DL (data loss) issues and code fixes for the raid related issues of Storage boxes. Also writing automated scripts to analyze the logs and find any known issues in the same.

Project Name:

Analysis and code fixes reported for Seagate storage boxes.

Domain: Storage

Environment: C, C++, Python

Description:

Analysis on logs for the issues reported by customer on storage issue and analyzes the data un-availability and data loss cases.

Working on raid & backend firmware stack to fix the possible code issues to avoid storage outages is any caused. Write scripts in Python to automate the logs processed from storage controllers and find any known issues in the same to speed up the analysis process.

Role & Responsibilities

- ⤴ Involved in adding new features and improvements in the RAID & Backend code.
- ⤴ Handling customer issues and providing the bug fixes.
- ⤴ Adding new scripts to analyze the logs automatically and report known issues.
- ⤴ Providing the root cause and avoiding the major DU & DL issues solving.

Projects at MSys Technologies

Senior Engineer – Dec 2016 – Nov 2017

Client: SEAGATE HDD India Pvt LTD

Working as Senior Engineer on Storage Products for DotHill products (Formally acquired by Seagate). Responsible in doing analysis to root cause the DU (data unavailability) DL (data loss) issues and code fixes for the raid related issues of Storage boxes.

Project Name:

Analysis and code fixes on DU, DL cases reported for Seagate storage boxes.

Domain: Storage

Environment: C, C++

Description:

Analysis on logs for the issues reported by customer on storage issue and analyzes the data un-availability and data loss cases. Working on raid firmware stack to fix the possible code issues to avoid storage outage and report the same to CS.

Role & Responsibilities

- ⤴ Involved in adding new features and improvements in the RAID code.
- ⤴ Handling customer issues and providing the bug fixes.
- ⤴ Providing the root cause and avoiding the major DU & DL issues solving.

Projects at Oracle

Software Engineer – Jul 2010 – Nov2016

Oracle India Pvt LTD Hyderabad, India

Working as Software Developer on Storage Products for Oracle Flash Storage systems & AXIOM 600/500.

Project Name:

RAID Firmware code enhancement for Flash Storage System.

Domain: Storage

Environment: C, C++

Description:

RAID firmware handles the drives connected to the enclosures providing virtualization to the upper layers (host) where RAID firmware is also responsible for IO transfer from disk drives to host. There are enclosures connected in array and each enclosure is having total 22 drives. The enclosures are connected in such a way that at least there is always one path available for the downstream enclosures. Here which Responsible for analysis, design, and coding the modules of Storage SAN products.

Role & Responsibilities

- ⤴ Involved in adding new features and improvements in the RAID code.
- ⤴ Handling internal bugs and providing fixes.
- ⤴ Handling Customer live brick recoveries.
- ⤴ Recovering customer data with customized firmware.(special code)

Module working: Partial Offline feature in FS release 6.

When a drive was failed due to write failure on one node but was not experienced the same issue, we keep the drive offline only to one node and make the RAID LUN online and available from companion node. This will make the array to be redundant even the drive was noticed fault on one node. We will perform rebuild once the drive was faulted on both the nodes only.

Working on Sync the nodes with the partition data and decide the further actions to be made to make the data availability.

Module Ownership: Copy away in release 4 & 5

When a drive in the array has tripped the threshold limit of the errors like SMART, Average execution time we believe this drive is bad but instead of failing it and making the lun degraded array. We try to initiate copy-away from this drive to the hot spare in the array.

Thus, we fault the source drive once all the data is migrated to spare disk. In some cases if the source drive fails in between the copy away process we clean up the target hot spare and will initiate rebuild.

I had worked in this area of code fixing very rare corner timing window issues for having rebuilt to progress smoothly after we fail a copy-away source drive.

Projects at Vedams Software Solutions

Software Engineer – Jul 2010 – Jul 2013

Vedams Software Solutions Pvt Ltd, Hyderabad, India

Working as Software Engineer on Storage Products for Pillar Data Systems (Formally acquired by Oracle). Responsible in doing analysis, design, and coding the modules of Storage SAN products.

Project Name:

RAID Firmware code enhancement for Axiom 500/600 System.

Domain: Storage

Environment: C, C++

Description:

Brick Firmware virtualizes the hard drives with RAID arrays. Bricks are available with SSD, FC, and SATA allowing multiple Brick types. Each SSD and SATA Brick supports 13 drives, each FC Brick will support up to 12 drives. Brick contains two RAID controllers; you can add more Bricks to expand your capacity. With the Axiom system you are able to optimize application performance based on application I/O operations.

Role & Responsibilities

- ⤴ Involved in adding new features and improvements in the RAID code.
- ⤴ Handling severity 1 (P1) issues and providing the bug fixes.
- ⤴ Handling Customer live data recoveries.

Project Name:

SAMBA CTDB Integration for Lustre File Systems(LFS).

Description:

Domain: Storage

Environment: C, Shell

The objective of this project is to allow the shares of the clustered file system i.e. Lustre to be exported to Windows/Linux clients via Clustered Samba Servers.

CTDB provides the same types of functions as TDB but in a clustered fashion, providing a TDB-style database that spans multiple physical hosts in a cluster. Samba, which is a software suite that provides seamless file and print services to SMB/SMB2 and these, are used to export Lustre file system to the windows machines.

Role & Responsibilities

Work on R&D to configure CTDB and Samba.

Successfully integrated SAMBA, CTDB so to work even when there is a node failure, providing seamless connectivity between lustre file system and windows client.

Experience Prior to Vedams:

Description: Worked as Assistant Professor in Electronics & Communication Department from Jun 2009 to May 2010
RVR Institute of Engineering & Technology, Sheriguda, Hyderabad.

Roles & Responsibilities:

Handling academic subjects for BE & MCA students

Subjects Handled: Embedded Systems for MCA and Microprocessor-Micro-controllers, Electronic Circuit Analysis for the BE academic students.

Training the students on various mini and major academic projects.

Extra-Circular Activities:

- ▲ Captain of college Kabbadi team
- ▲ Member of an RED CROSS Committee.
- ▲ Member of an JEVAN blood bank.
- ▲ Doing Mini R&D projects
- ▲ Teaching/helping students on electronic projects.

Personal Profile:

Name	Gudala Manoj Kumar
Father's name	G Masthan Reddy
Mother's name	G Sarawathi
Date of birth	26 th July 1984
Nationality	Indian
Passport number	M6044514
Passport issue & expiry	2015 - 2025

References:

Available upon request